Human Lumbar Vertebrae
Quantitative Three-Dimensional Anatomy

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This study details the quantitative three-dimensional surface anatomy of human lumbar vertebrae based on a study of 60 vertebrae. The two lower vertebrae (L4 and L5) appeared to be transitional toward the sacral region, whereas the upper two vertebrae (L1 and L2) were transitional toward the thoracic region. Means and standard errors of the means for linear, angular, and area dimensions of vertebral bodies, spinal canal, pedicle, pars interarticularis, spinous and transverse processes were obtained for all lumbar vertebrae. This information provides a better understanding of the spine, and allows for a more precise clinical diagnosis and surgical management of spinal problems. The information is also necessary for constructing accurate mathematical models of the human spine. [Key words: lumbar spine, vertebrae, anatomy, vertebral dimensions, spinal canal, pedicles]

THE SHAPES AND SIZES of lumbar vertebrae are often given in artist's drawings based on visual inspection of cadavers,[6,7] with no quantitative geometric information provided. The quantification of lumbar vertebrae into a geometric data base has several applications. Physical properties of vertebrae are related to their size and shape. The size of the spinal canal has been found to be related to the incidence of low-back pain,[14] and the size is affected by physiologic stresses in childhood.[2] The shape of a lumbar vertebra—specifically, the concavity in the horizontal plane of its posterior wall—has been postulated to affect the frequency of disc herniation. The size and orientation of pedicles are important dimensions for the design of transpedicular fixation devices as well as associated surgical procedures. Several studies have been conducted in the past few years with the primary goal of documenting the pedicle dimensions for internal fixation device application.[8,9,17] These studies, however, documented only the geometric parameters relevant to pedicle devices, and not all the vertebral dimensions. Other recent studies by Berry et al[1] and Scoles et al[15] documented the anatomy of more vertebral elements.

Another important use for quantified anatomy is the development of accurate and valid mathematical models of the human spine. Mathematical models depicting the lumbar spine have been used to study the mechanical response of the spine to external forces,[16] and these responses with accompanying spinal instrumentation.[5] Currently, the geometric data concerning vertebrae used in these spinal models are generally of much lower quality than the mathematical sophistication of the models. As the quality of the model's prediction depends on the quality of the incorporated data, it is essential that high-quality data sets be available.

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Supported in part by National Institute of Health Grant AR39209.
Accepted for publication April 19, 1991.

The purpose of this paper is to report quantitative three-dimensional surface geometry of the lumbar spine from the L1 to L5 vertebrae. This is the final paper in a series of three, the others having covered the cervical[12] and thoracic vertebrae.[13]

MATERIALS and METHODS

As this report is the third in a series of anatomic studies, a detailed description of the methods can be found in the initial document by Panjabi et al.[13] A brief description follows.

Twelve fresh autopsy spine specimens (axis to last lumbar vertebra) were obtained and radiographed to identify any significant abnormalities. The specimens were identified by age, sex, and height. The subjects had an average age of 46.3 (range, 19–59) years, weight of 67.8 (range, 54–85) kg, height of 167.8 (range, 157–178) cm, and the male/female ratio was 8/4. Each specimen was prepared by completely removing all soft tissues from the vertebrae. Two pointed threaded rods (2 mm diameter and 2.5 mm long) were secured into each vertebral body. The two ends of one rod constituted two marker points, whereas the third marker point was provided by the end of the second threaded rod. Anteroposterior and lateral radiographs were taken to identify the pins in the vertebral body coordinate system, with the origin at the center of the upper end plate. A lumbar vertebra with the coordinate system seen in three views is shown in Figure 1.

Linear dimensions, angulations, and areas of various vertebral parts were studied. A summary of all measured dimensions is listed in Table 1 and shown graphically in Figure 1. Each vertebral part was defined in space by a series of points spanning or circumventing its surface. For example, each articular facet was defined by points (up to 20 points) spanning its entire surface. Conversely, each end plate, spinous process, transverse process, and pedicle was defined by points circumventing the structure. The number of points chosen was such that the geometry was well defined by straight lines joining the adjacent points.

The three-dimensional coordinates of all the points with respect to a reference system defined by the three marker points were determined using a specially designed morphometer.[10] This instrument consisted of one linear (LVDT) and two rotational (RVDT) displacement transducers arranged such that their axes meet at one point, thus establishing a spherical coordinate system (with coordinates r, θ, ϕ). The location of any point, or series of points, with respect to the three marker points, was determined by placing the tip of the morphometer at that point(s) in question and at the three marker points. The three-dimensional coordinates were recorded directly into the computer. In this manner, the morphometer was used to record the three-dimensional positions of all points that defined the vertebral parts listed in Table 1. In addition to the parts listed in Table 1, four points defining the left and right lower and upper edges of the posterior wall of the vertebral body were also digitized. These four points were used to define the vertebral body reference planes: frontal (xy), transverse (xz), and sagittal (yz), using a least-squares technique. The position of the center of the upper end plate was digitized as the origin for the vertebral body coordinate system (Figure 1).

Three types of anatomic parameters were computed, and are described below.

Linear Parameters. Computer software was developed to calculate the various linear parameters from the three-dimensional digitized

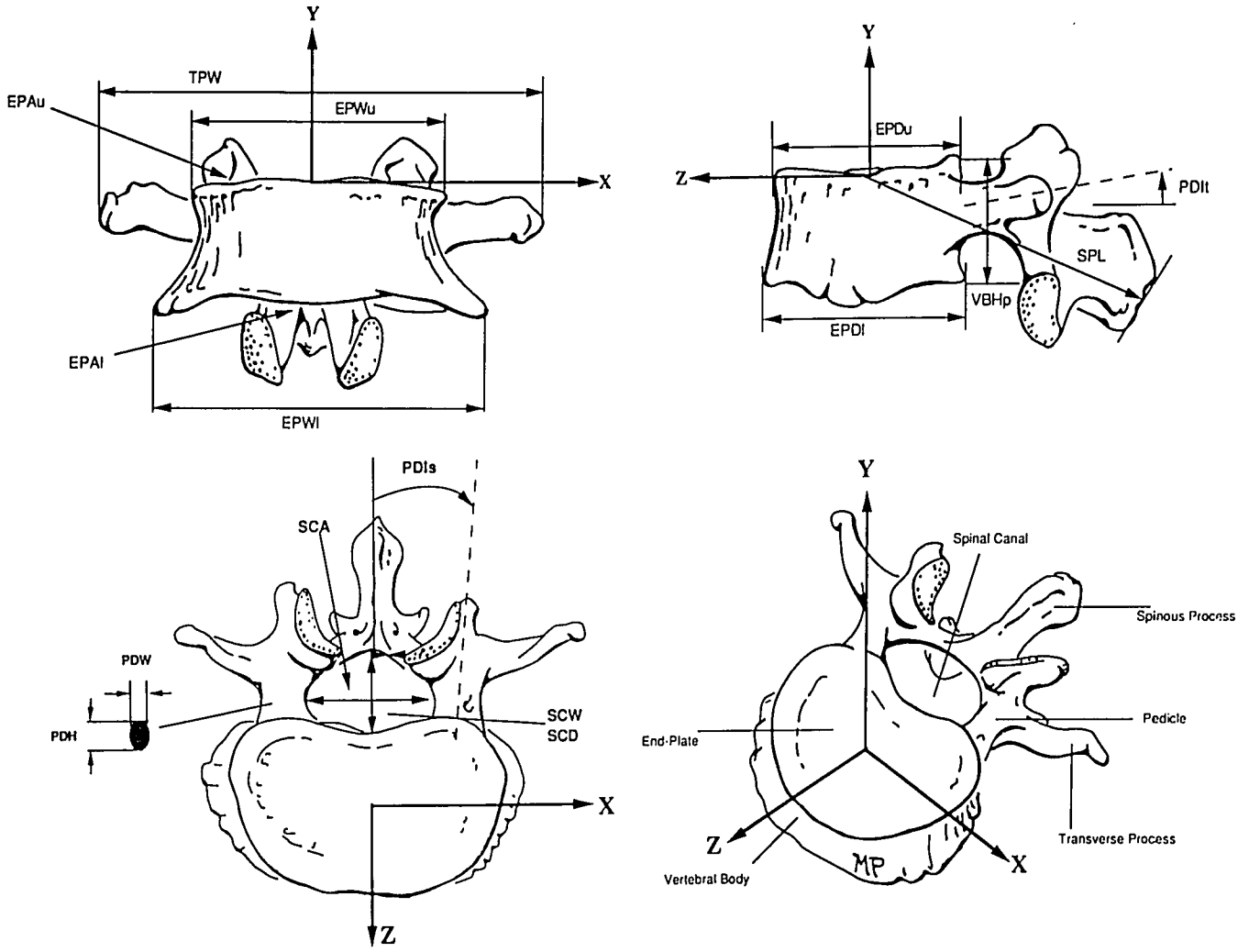


Fig 1. Three orthogonal views (front, side, and top) of a lumbar vertebra are shown. The various dimensions determined in the current study are indicated, and are further explained in the nomenclature given in Table 1. The fourth view is an isometric view showing the coordinate system used.

coordinates. The major and minor linear dimensions of each vertebral part were calculated. For example, each pedicle was described by its width and height. In addition, various gross vertebral body dimensions were calculated, such as the height of the vertebral body and the distances between transverse processes. The results of 12 specimens were pooled to calculate mean and standard error of the mean for each linear dimension.

Surface and Cross-Sectional Area Parameters. The projected areas of the digitized parts onto the frontal (xy), sagittal (yz), and transverse (zx) planes were computed by considering only two of the three coordinates of each point constituting the part. For example, to calculate frontal plane area, only the x and y coordinates of the points would be considered. The total area was computed as the Euclidean (ie, square root of the sum of the individual squares) sum of the three projected areas. The means and standard error of the means of all specimens were calculated.

Angular Parameters. To calculate angulations, the surface of each vertebral part was represented by a plane. Since each part was characterized by more than three points, the equation of a plane was used to yield the plane that would represent the best fit to the surface, using the least-squares technique. The angles between the best fit plane and each anatomic plane were obtained.

Statistical Analysis. For all parameters, differences between vertebral levels were determined using a single factor analysis of variance (ANOVA) and Scheffe's F test for pairwise comparison at a 95% significance level.

Accuracy of Measurements. The accuracy in computing linear, angular, and surface area parameters was estimated using an object with known values. A Plexiglas piece with sides cut at 30°, 45°, and 60° was used. A sheet of millimeter graph paper was pasted on each of the cut surfaces, and different shapes (triangles, rectangles, circles, etc.) were drawn on it. The three marker points, similar to those placed on a vertebra, were identified on the Plexiglas piece. Thus, a pseudo-vertebra with surfaces of known inclinations and parts of known dimensions and areas was created. The parts were digitized with the same protocol and computer programs adopted for the vertebral parts. The linear, angular, and surface area parameters were computed and compared with the known values. The error was found to be less than 5% in all cases.

RESULTS

A total of 12 C2 to L5 spines was studied. This report describes results from the 60 lumbar vertebrae that were examined and measured using the above protocol. As described, the quantification of vertebral

Table 1. Table of Nomenclature (Refer to Figure 1)

	Mnemonics	*Dimension*
Body part		
Vertebral body	EPW	End-plate width
	EPD	End-plate depth
	EPA	End-plate area
	EPI	End-plate inclination
	VBH	Vertebral body height
Spinal canal	SCW	Spinal canal width
	SCD	Spinal canal depth
	SCA	Spinal canal area
Pedicle	PDW	Pedicle width
	PDH	Pedicle height
	PDA	Pedicle area
	PDI	Pedicle inclination
Spinous process	SPL	Spinous-process length
Transverse process	TPW	Transverse process width
Pars interarticularis	PIA	Pars interarticularis area
Suffixes	a	Anterior
	p	Posterior
	s	Sagittal plane
	t	Transverse plane
	f	Frontal plane
	u	Upper
	l	Lower (or left, depending on context)
	r	Right

Note: The mnemonics for each body part consist of three upper case letters, the first two representing the part and the third representing the measured dimension. This three-letter combination is followed by lower case characters (suffixes) indicating direction, plane, left/right, etc.

parts was done in terms of three types of parameters: linear, angular, and surface area dimensions. The complete results of the parameters measured for the lumbar vertebrae (L1–L5) are listed in Tables 2, 3, 4, and 5.

The nomenclature for the measured parameters is presented graphically in Figure 1, and also given in Table 1. The results are shown graphically as well as described below, whereas statistical differences are shown in the tabulated data.

Vertebral Body

The linear dimensions of the upper and lower end plates consisted of lateral width (EPW) and anteroposterior depth (EPD). There was a slight increase from L1 to L5 in the end-plate widths of approximately 14% (Figure 2A). The end-plate depths remained relatively constant throughout the lumbar spine.

The ratio of the width to the depth increased from L1 to L5 for both the upper and lower end plates (Figure 2B). For the upper end plate, the increase was approximately 12% of the value at L1, whereas the lower end-plate ratio increased by 21% from L1 to L5.

In relation to the total area, the upper and lower end plates at L5 were respectively 17% and 9% larger than those at L1. The increase was steady and essentially linear from L1 to L4 for the upper end plate, whereas for the lower end plate, L3 had the greatest area (Figure 2C). Using the end-plate linear dimensions of width and depth, its area was estimated as an ellipse ($A = \pi \times EPW \times EPD/4$). This curve (dotted line) is shown in Figure 2C.

Spinal Canal

The linear dimensions of the spinal canal were lateral width (SCW) and anteroposterior depth (SCD). The lateral width increased steadily from L1 to L5. The curve for the spinal canal depth resembled a parabola, decreasing from L1 to L3 and then increasing from L3 to L5. The narrowest portion in the lateral dimension (SCD) occurred at L3, with an average value of 17.5 mm.

The ratio between the width and depth is shown in Figure 3B. This ratio increased from L1 to L3, and remained constant through to L5.

The cross-sectional area of the spinal canal decreased from L1 to L2, constant from L2 to L4, and then increased to L5 (Figure 3C). As with the end plates, an attempt was made to model the spinal canal area both as an ellipse and a triangle. The results are shown in Figure 3C.

Pedicle

The linear dimensions of the pedicle were its cross-sectional height (PDH) and width (PDW). The height (PDH) averaged 15.9 mm at L1, decreased to 14.4 mm at L3, and increased to 19.6 mm at L5 (Figure 4A). The width (PDW) was initially 8.6 mm at L1, then decreased to 8.2 mm at L2. From L2 to L5, there was an increase of approximately 3.5 mm per vertebral level. At L5, the width was 18.9 mm.

The height to width ratios for the left, right, and average pedicles are

Table 2. Vertebral Body Dimensions

	Lumbar Vertebral Level				
	L1	*L2*	*L3*	*L4*	*L5*
Linear dimensions (mm)					
EPDu	34.1 ± 1.34	34.6 ± 1.10	35.2 ± 1.10	35.5 ± 0.88	34.7 ± 1.17
EPWu	41.2 ± 1.03	42.6 ± 0.74	44.1 ± 0.88	46.6† (L1) ± 1.20	47.3† (L1) ± 1.20
EPDl	35.3 ± 1.27	34.9 ± 0.74	34.8 ± 1.24	33.9 ± 0.85	33.2 ± 0.92
EPWl	43.3 ± 0.78	45.5 ± 1.10	48.0 ± 1.24	49.5† (L1) ± 1.38	49.4† (L1) ± 1.41
VBHp	23.8 ± 1.03	24.3 ± 0.95	23.8 ± 1.10	24.1 ± 1.10	22.9 ± 0.95
EPWu/EPDu	1.21	1.23	1.25	1.31	1.36
EPWl/EPDl	1.23	1.30	1.38	1.46	1.49
Surface Areas (mm^2)					
EPAu	1057 ± 60.78	1136 ± 61.84	1194 ± 55.19	1239 ± 58.41	1237 ± 58.48
EPAl	1117 ± 49.00	1197 ± 51.41	1290 ± 64.35	1273 ± 51.72	1218 ± 59.43
Angular dimensions (°)					
EPltu	2.7 ± 0.77	3.5 ± 1.5	1.7 ± 0.43	4.7 ± 1.1	2.2 ± 0.54
EPltl	4.0 ± 0.84	2.1 ± 0.4	2.7 ± 1.1	2.7 ± 0.69	1.8 ± 0.42

*Mean ± SEM.
†$P < .05$ (comparison levels shown in parentheses).

Table 3. Spinal Canal Dimensions

	Lumbar Vertebral Level				
	L1	*L2*	*L3*	*L4*	*L5*
Linear dimensions (mm)					
SCW	23.7 ± 0.92*	23.8 ± 0.71	24.3 ± 0.64	25.4 ± 0.49	27.1 ± 0.88
SCD	19.0 ± 0.67	18.2 ± 0.53	17.5 ± 0.53	18.6 ± 0.71	19.7 ± 0.49
SCW/SCD	1.25	1.31	1.39	1.37	1.38
Cross-sectional areas (mm^2)					
SCA	320 ± 18.10	281 ± 15.38	280 ± 14.64	290 ± 15.17	330 ± 21.21

*Mean ± SEM.
Note: For each parameter, an analysis of variance (ANOVA) was performed to determine which vertebral levels were different from one level to the other. For nomenclature, see Table 1 and Figure 1.

Table 4. Pedicle Dimensions

	Lumbar Vertebral Level				
	L1	*L2*	*L3*	*L4*	*L5*
Linear dimensions (mm)					
PDWr	8.0 ± 0.95*	7.8 ± 0.57	10.2 ± 0.67	13.4† (L1-L2) ± 0.18	18.0† (L1-L4) ± 1.03
PDHr	15.9 ± 0.81	15.0 ± 0.53	14.2 ± 0.64	15.7 ± 0.57	19.6† (L1-L4) ± 0.74
PDWl	9.2 ± 0.88	8.7 ± 0.78	10.1 ± 0.53	14.7† (L1-L3) ± 0.46	19.2† (L1-L4) ± 1.03
PDHl	15.8 ± 0.74	14.9 ± 0.53	14.6 ± 0.60	15.2 ± 0.46	19.5† (L1-L4) ± 0.99
PDHr/PDWr	2.00	1.91	1.39	1.17	1.09
PDHl/PDWl	1.72	1.71	1.44	1.03	1.02
Cross-sectional areas (mm^2)					
PDAr	86.4 ± 11.42	83.3 ± 8.94	92.9 ± 7.71	102.8 ± 6.65	143.9† (L1-L3) ± 9.55
PDAl	88.5 ± 11.74	83.6 ± 8.87	95.3 ± 9.37	112.3 ± 7.53	158.7† (L1-L4) ± 10.82
Angular dimensions (°)					
PDIsr	16.5 ± 5.02	17.1 ± 3.75	19.8 ± 2.33	18.4 ± 1.66	25.9 ± 1.73
PDItr	2.2 ± 0.67	3.3 ± 0.67	2.9 ± 1.06	4.8 ± 1.06	5.2 ± 1.80
PDIsl	12.4 ± 1.87	11.2 ± 2.02	17.1 ± 1.56	14.7 ± 2.16	23.2† (L1-L2) ± 2.51
PDItl	2.9 ± 0.74	2.1 ± 0.64	2.4 ± 0.67	3.0 ± 1.17	5.7 ± 1.48

*Mean ± SEM.
†$P < .05$ (comparison levels shown in parentheses).
Note: For each parameter, an analysis of variance (ANOVA) was performed to determine which vertebral levels were different from one level to the other. For nomenclature, see Table 1 and Figure 1.

shown in Figure 4B. At L1 and L2, the average ratio was 1.84. With progression down the lumbar spine, the ratio decreased to approximately 1.0 at L5.

The cross-sectional areas of both the left and right pedicles from L1 to L5 are shown in Figure 4C. Initially, there was a decrease from L1 to L2, followed by a relatively constant increase to L5. The pedicle shape was estimated as an ellipse ($\pi \times PDW \times PDH/4$) and a triangle ($PDW \times PDH/2$), as shown in Figure 4C, to determine which geometric shape best approximated the pedicle area.

The pedicle angle was defined as that of its center line (perpendicular to the pedicle cross-section) with respect to each of the two planes: sagittal and transverse. The inclination measurements of the pedicles showed a general left/right symmetry, as seen in Figure 4D. The progression of inclination relative to the sagittal plane (PDIs) increased generally from approximately 14.5° at L1 to 24.5° at L5. The inclination with the transverse plane (PDIt) increased slightly from 2.5° at L1 to 5.5° at L5.

Using the measured height and width of each pedicle, the second area moments of inertia with respect to its axes were computed ($PDMs = \pi \times PDH^3 \times PDW/64$, $PDM = \pi \times PDH \times PDW^3/64$), and are shown in Figure 4E. These values are proportional to the bending stiffness of the pedicle in the sagittal and transverse (horizontal) planes, respectively. As shown in Figure 4E, each moment of inertia remained relatively constant from L1 to L3, and rose sharply to L5. The sagittal plane stiffness was always larger than the transverse plane stiffness.

Pars Interarticularis

The pars interarticularis area (PIA) showed a gradual increase from L1 to L5 (Figure 5, Table 5). The cross-sectional area at L1 was about 62 mm^2, and at L5, it was 81 mm^2.

Spinous and Transverse Processes

The spinous process length (SPL) was measured from the center of the superior end plate (origin of the coordinate system) to the most inferoposterior tip. At L1, it was about 68 mm (Figure 6, Table 6). The length of the process increased slightly to 72 mm at L3, and then decreased to 68 mm at L5.

The transverse process width (TPW) was defined as the distance from the tip of the left transverse process to that of the right (Figure 6, Table 6). It increased generally from 71 mm at L1 to 92 mm at L5.

DISCUSSION

To our knowledge, the current study is unique in that it provides, for the first time, a comprehensive quantitative three-dimensional surface anatomy of the lumbar vertebrae. Problems of spinal instability, spinal trauma, and, more recently, the design of pedicle screw fixation devices

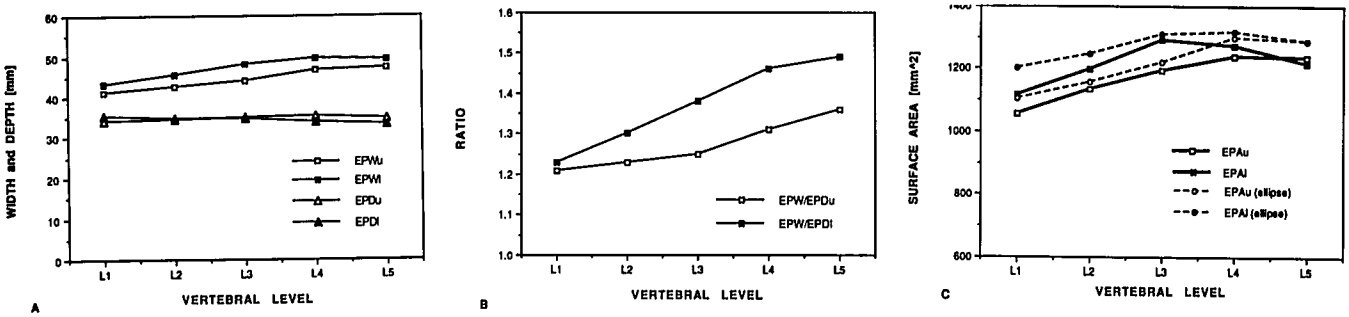


Fig 2. Vertebral end-plate dimensions as functions of vertebral levels L1 to L5 are shown. **A,** Linear dimensions of the upper and lower end plates. **B,** Ratio of end-plate width to depth for the upper and lower end plates. **C,** Surface areas of upper and lower end plates are indicated by whole lines. Dashed lines depict the end-plate areas, assuming the end-plate shapes to be an ellipse.

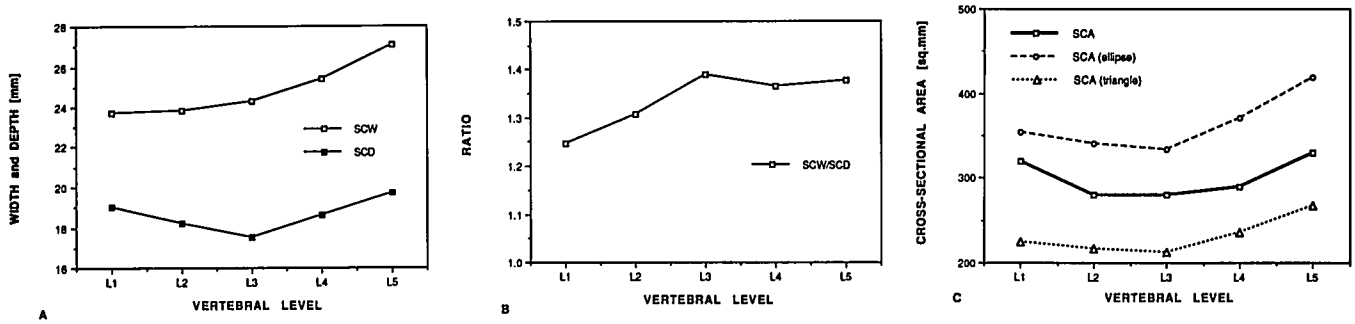


Fig 3. Spinal canal dimensions as functions of vertebral levels L1 to L5 are shown. **A,** Spinal canal width and depth. **B,** The ratio of spinal canal width to depth. **C,** Cross-sectional area of the spinal canal is shown by whole lines. Dashed lines indicate the spinal canal areas, assuming its shape to be either an ellipse or a triangle.

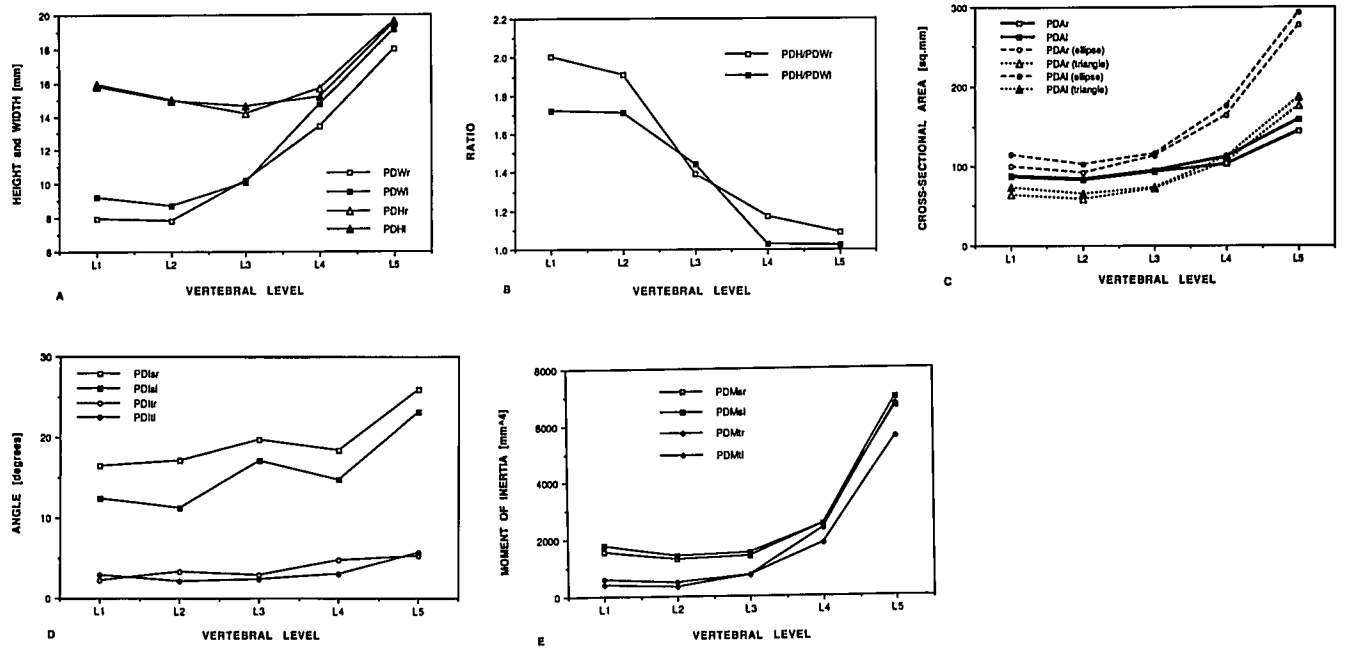


Fig 4. Pedicle dimensions as functions of vertebral levels L1–L5 are shown. **A,** Widths and heights of the pedicle for right and left. **B,** The ratio of height to width for the right and left pedicles. **C,** Cross-sectional areas of the left and right pedicles are shown by whole lines. The dashed lines indicate the areas, assuming these to be either an ellipse or a triangle. **D,** Pedicle axis angulations with respect to the sagittal and transverse planes for the right and left pedicles. **E,** Pedicle moments of inertia with respect to the sagittal and transverse planes, calculated from the cross-sectional height and width.

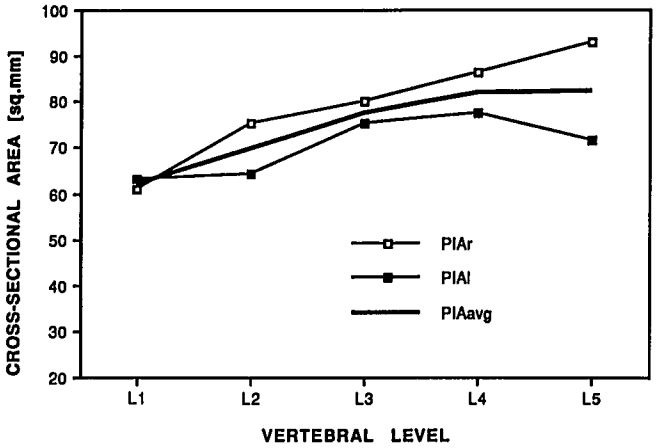


Fig 5. Left, right, and average pars interarticularis areas as functions of vertebral levels L1–L5.

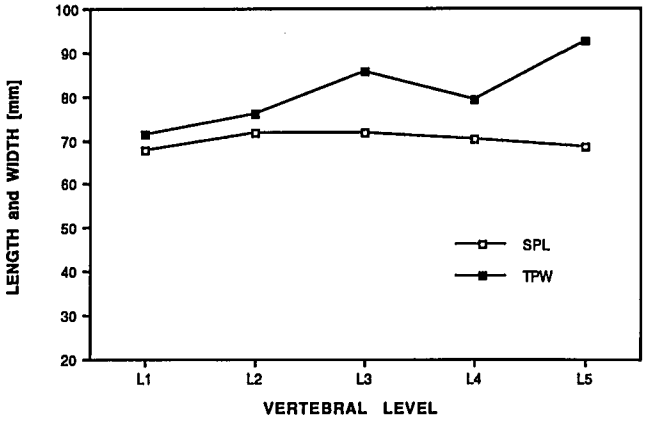


Fig 6. Spinous and transverse process dimensions as functions of vertebral levels L1–L5.

are some clinical examples of areas that may benefit from this quantified anatomic data base, through the use of more precise dimensions of the spinal canal, and the development of more anatomically accurate mathematical models. Additionally, the quantified three-dimensional data presented here are essential for the construction of accurate spinal mathematical models.

Our methodology incorporated relatively simple and inexpensive equipment, in contrast to much more expensive techniques, such as computed tomography scans, and achieved a high degree of accuracy.

Transition Zones

From the data presented here, the lumbar spine may be naturally divided into three distinct regions: the transition from thoracic to lumbar; the transition from lumbar to sacral; and a middle region. It may be of interest to mention that the coupling patterns of motions in the lumbar spine are also different in these three regions.[11]

The upper region, from thoracic to lumbar, was characterized best by a narrowing of the spinal canal depth from L1 to L3, whereas the widths remained relatively constant. The decrease in spinal canal depth contrasted to its increase from T10 to T12.[13] As a result, the width to depth ratios increased from L1 to L3, thus identifying this upper transition zone. Also in this region, the pedicle heights decreased slightly from L1 to L3, a continuing trend from T11 to T12.[13] The pedicle widths were very small, only 8–9 mm, in contrast to 15–16 mm heights at the lower lumbar levels.

The middle lumbar region was characterized by a relatively narrow spinal canal at L3. In addition, this middle lumbar region was characterized by the greatest end-plate areas and the longest spinous process lengths, probably results of the natural lordotic curve reaching its apex at the L3 vertebra.

The final transition zone, from lumbar to sacral, was most evident by a distinct enlarging of the spinal canal area from L3 to L5. From L3 to L5, the pedicles increased dramatically in size, due to the increasingly large loads carried by the lower lumbar levels. This is substantiated by the large increase in pedicle moments of inertia. At L5, the pedicle angle with respect to the sagittal plane increased by an average of 8° over the value at L4. This fanning out represents a distinct transition to the sacrum.

Previous Studies

Several quantitative anatomy studies have been completed recently for the lumbar vertebrae.[1,3,4,8,9,15,17] In three of the studies,[8,9,17] the focus lay solely on the pedicle dimensions that are clinically important for design and surgical implantation of internal fixation devices. Table 6 provides a comparison between the various parameters measured in the current study and some of the earlier work.

The first parameter that we compared was the anteroposterior depth of the lower end plate (EPDl). The values corresponded well to those of the other studies. The lower end-plate widths (EPWl) agreed best with the results of Scoles et al,[15] but were generally several millimeters less than the data of Berry et al,[1] and larger than those of Cotterill et al.[3] The posterior vertebral body heights (VBHp) reported herein were very close to those given by Cotterill et al, and were consistently 1–2 mm less than those reported by Berry et al, Scoles et al, and Gilad and Nissan.[4]

The spinal canal depths (SCD) in the current study were slightly larger than the data of Berry et al and Scoles et al, but each study displayed the same trends throughout the lumbar spine. The spinal canal

Table 5. Pars Interarticularis, Spinous Process, and Transverse Process

	Lumbar Vertebral Level				
	L1	*L2*	*L3*	*L4*	*L5*
Cross-sectional area (mm^2)					
PIAr	61.0 ± 2.19*	75.2 ± 6.58	70.0 ± 6.86	86.2 ± 7.39	92.9† (L1) ± 9.76
PIAl	63.2 ± 1.17	64.4 ± 5.90	75.1 ± 8.91	77.5 ± 8.91	71.7 ± 9.69
Linear dimensions (mm)					
SPL	67.7 ± 1.24	71.7 ± 1.70	71.7 ± 1.77	70.1 ± 1.17	68.3 ± 1.63
TPW	71.2 ± 1.66	76.1 ± 1.48	85.7† (L1) ± 2.44	79.4 ± 3.78	92.5† (L1,L2,L4) ± 3.68

*Mean ± SEM.

†$P < .05$ (comparison levels shown in parentheses).

Note: For each parameter, an analysis of variance (ANOVA) was performed to determine which vertebral levels were different from one level to the other. For nomenclature, see Table 1 and Figure 1.

Table 6. Comparison of Results of Present Study to Those of Previous Studies*

Parameter	Level	Present Study	Berry (1987)	Zindrick (1987)	Scoles (1988)	Krag (1986, 1988)	Cotterill (1986)	Gilad (1986)
Linear dimensions (mm)								
EPDI	L1	35.3	32.3	—	29.5	31.0	—	34.1
	L2	34.9	33.4	—	—	32.5	—	34.7
	L3	34.8	34.2	—	32.6	30.5	32.7	34.6
	L4	33.9	35.6	—	—	32.0	—	34.9
	L5	33.2	34.5	—	34.5	33.0	—	33.9
EPWI	L1	43.3	49.1	—	44.3	—	—	—
	L2	45.5	54.8	—	—	—	—	—
	L3	48.0	53.8	—	48.4	—	44.3	—
	L4	49.5	50.9	—	—	—	—	—
	L5	49.4	52.7	—	52.9	—	—	—
VBHp	L1	23.8	25.8	—	27.4	—	—	27.1
	L2	24.3	25.2	—	—	—	—	28.0
	L3	23.8	26.0	—	27.4	—	23.0	27.9
	L4	24.1	26.4	—	—	—	—	27.1
	L5	22.9	23.1	—	27.6	—	—	25.7
SCD	L1	19.0	17.2	—	17.6	—	—	—
	L2	18.2	16.0	—	—	—	—	—
	L3	17.5	16.2	—	16.0	—	12.1	—
	L4	18.6	16.1	—	—	—	—	—
	L5	19.7	17.3	—	17.6	—	—	—
SCW	L1	23.7	22.1	—	22.2	—	—	—
	L2	23.8	23.0	—	—	—	—	—
	L3	24.3	22.7	—	22.8	—	21.5	—
	L4	25.4	22.0	—	—	—	—	—
	L5	27.1	26.0	—	25.9	—	—	—
PDH	L1	15.9	15.6	15.4	15.3	—	—	—
(avg.	L2	14.9	15.3	15.0	—	—	—	—
left and	L3	14.4	14.5	14.9	14.1	—	—	—
right)	L4	15.4	13.1	14.8	—	—	—	—
	L5	19.6	13.7	14.0	16.2	—	—	—
PDW	L1	8.6	7.0	8.7	8.3	7.0	—	—
(avg.	L2	8.3	7.5	8.9	—	8.4	—	—
left and	L3	10.2	9.2	10.3	9.1	9.1	8.4	—
right)	L4	14.1	10.4	12.9	—	10.8	—	—
	L5	18.6	10.7	18.0	9.7	13.8	—	—
Angular dimension (°)								
PDIs	L1	14.5	7.5	10.9	11.6	11.0	—	—
	L2	14.1	11.5	12.0	—	12.0	—	—
	L3	18.5	14.0	14.4	14.7	16.0	—	—
	L4	16.0	20.0	17.7	—	18.0	—	—
	L5	24.6	31.5	29.8	23.1	27.0	—	—
PDIt	L1	2.6	—	2.4	—	—	—	—
	L2	2.7	—	1.8	—	—	—	—
	L3	2.7	—	0.2	—	—	—	—
	L4	3.9	—	0.2	—	—	—	—
	L5	5.5	—	−1.8	—	—	—	—

*The average values of selected linear and angular dimensions are compared.
Note: For nomenclature, see Table 1 and Figure 1.

widths (SCW) of all studies agreed extremely well in both magnitude and trend.

The pedicle heights (PDH) of this study agreed well with the results of Berry et al, Zindrick et al,[17] and Scoles et al for the upper lumbar levels. However, Berry et al and Zindrick et al found the heights to decrease at L4 and L5, whereas our study and Scoles et al found large increases towards the lower lumbar levels. The average pedicle widths (PDW) in this study agreed with the data of Zindrick et al at all levels. However, Berry et al, Scoles et al, and Krag et al[8] found small increases at L5. There is remarkable consistency between all studies in the angulation of the pedicles with respect to both the sagittal (PDIs) and transverse (PDIt) planes.

Geometric Shape Approximations

As an aid to mathematical modeling of the spine, it is useful to be able to approximate various vertebral components by simple geometric shapes. In this study, the surface areas of the inferior and superior end plates were approximated as ellipses, while the cross-sectional areas of the spinal canal and pedicles were approximated by ellipses and triangles based on their measured widths and depths or heights.

For both the lower and upper end plates, the elliptical approximation overestimated the measured area by less than 10%.

For the spinal canal, the elliptical approximation was a consistent overestimate by 10–30%, whereas the triangular approximation underestimated the actual area by 20–30%. The elliptical approximation was better at the upper lumbar levels, whereas the triangular estimate was best at the lower levels.

For the pedicle cross-section, the triangular approximation was best for levels L4 and L5, whereas the elliptical estimate was best at L1 and L2. At L3, the measured areas were about midway between the triangular and elliptical approximations.

SUMMARY

A comprehensive data set has been presented that provides quantitative anatomy of lumbar vertebrae. We have attempted to make a "blueprint" of the external geometry of lumbar vertebrae. Such infor-

mation is essential for the development of advanced and valid mathematical models of the spine. The latter may be used to predict spine behavior (eg, effects of surgery, injury mechanisms). The data may also be useful directly in clinical situations, such as in instability determination and the design of internal spinal devices, both functional and fixation types.

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